Supplementary material

Interpretable Prediction of Reported Older Person's Grant Non-Receipt Among Apparently Eligible Adults in KwaZulu-Natal

This supplement documents screening alternatives, model-validation sensitivities, descriptive statistics and the recorded literature searches. Tables S1–S4 use retained aggregate outputs; Table S5 reproduces the author's search record. Full-precision computational results remain in the cited CSV files. No respondent identifiers or new primary estimates are included.

The computational analyses use rule version opg-kzn-primary-v1 and primary seed 201415. The frozen cohort's archived CRLF representation has SHA-256 50c73c5ca88592dc426ae79f708dcc18cb057832fd2e00174a0257b4b1f6adc2; the equivalent LF workspace representation has SHA-256 6edcc6e93eda3f0c58243a64ab3281513abc7a5222f2b2f8836d373c98cf4625. AUC denotes area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large; and OOF, out-of-fold prediction. N and events count sampled adults and reported non-recipients. Household counts are distinct within each scenario. Displayed values are rounded.

Supplementary Table S1. Screening and model sensitivity scenarios

Panel A. Operational screening rules

Each row changes the stated component of S0 while retaining the remaining research rules. These alternatives test measurement assumptions; they are not complete determinations of statutory entitlement. Changed cohorts were refitted using five household-grouped folds and training-only imputation. Unchanged cohorts reused the primary predictions. All eight changed-cohort screening scenarios converged in five of five folds. Reused rows represent no additional fit. S1b and S1d selected the same additional people, although their rules differ.

ID	Change relative to S0	Prediction source
S0	Interview-month single/couple ceilings; nominal private-income proxy excludes social assistance; conservative spouse linkage; own disability or war-veterans receipt fails; resolved current OPG receipt defines the outcome.	Primary
S1a	Apply R61,800/R123,600 at every interview.	Reused
S1b	Apply R64,680/R129,360 at every interview.	Refit
S1c	For an applicant aged 75+, add R960 to the single ceiling and double that ceiling for couples (+R1,920). This applicant-age simplification does not model spouse-specific age supplements.	Reused
S1d	Use zero-payment single/couple boundaries R64,800/R129,600 before April 2015 and R67,680/R135,360 thereafter; equality fails. S0 instead uses the R1,200 annual minimum-payment boundary.	Refit
S2a	Treat living together (marital code 2) as a couple when a resident partner can be linked; retain spouse-link and resource checks.	Refit
S2b	Treat living together as single.	Refit
S2c	Treat separated adults (marital code 5) as single.	Refit
S2d	Treat both living together and separated as single.	Refit
S3	Retain own disability/war-veterans grant recipients. No war-veterans recipient was added in these data.	Refit
S4a	Replace nominal annualised income with the released CPI-adjusted values while retaining nominal ceilings. This deliberately mismatches price bases.	Reused
S4b	Keep social-assistance income in the resource proxy.	Refit
S4c	Also exclude workmen’s compensation, UIF and similar-fund income (COICOP 50333600).	Reused
S5	If current receipt disagrees with preceding-year OPG income, use the income record while retaining the eligibility screen. No such disagreement occurred among PASS adults.	Reused
S6a	Assign unresolved receipt after PASS to receipt. No such outcome occurred.	Reused
S6b	Assign unresolved receipt after PASS to non-receipt. No such outcome occurred.	Reused

Panel B. Cohort size and reported non-receipt

Changes are relative to S0. Unweighted percentages are events divided by N; person-weighted percentages divide the sum of non-recipient weights by the sum of all weights within the scenario’s resolved PASS cohort.

ID	Households	N	Events	ΔN	Δevents	Unw. %	Person-weighted %
S0	827	1,013	67	0	0	6.61	7.49
S1a	827	1,013	67	0	0	6.61	7.49
S1b	829	1,015	69	+2	+2	6.80	7.68
S1c	827	1,013	67	0	0	6.61	7.49
S1d	829	1,015	69	+2	+2	6.80	7.68

ID	Households	N	Events	ΔN	Δevents	Unw. %	Person-weighted %
S2a	870	1,072	71	+59	+4	6.62	7.42
S2b	871	1,073	69	+60	+2	6.43	7.17
S2c	831	1,021	67	+8	0	6.56	7.43
S2d	875	1,081	69	+68	+2	6.38	7.12
S3	840	1,032	82	+19	+15	7.95	9.06
S4a	827	1,013	67	0	0	6.61	7.49
S4b	822	1,007	67	−6	0	6.65	7.55
S4c	827	1,013	67	0	0	6.61	7.49
S5	827	1,013	67	0	0	6.61	7.49
S6a	827	1,013	67	0	0	6.61	7.49
S6b	827	1,013	67	0	0	6.61	7.49

Panel C. Scenario-specific held-out performance

Performance is unweighted. Changed cohorts can change case mix, event frequency and fold membership; differences therefore do not isolate a screening rule's effect on predictive quality. Matching values for unchanged cohorts arise from reusing the primary predictions, not independent replications.

ID	AUC	AP	Brier	CITL	Calibration slope
S0	0.752	0.203	0.0578	0.020	0.836
S1a	0.752	0.203	0.0578	0.020	0.836
S1b	0.763	0.211	0.0584	0.012	0.875
S1c	0.752	0.203	0.0578	0.020	0.836
S1d	0.763	0.211	0.0584	0.012	0.875
S2a	0.749	0.174	0.0588	−0.012	0.779
S2b	0.756	0.180	0.0564	0.008	0.841
S2c	0.753	0.205	0.0571	0.002	0.839
S2d	0.751	0.183	0.0565	0.002	0.808
S3	0.736	0.224	0.0682	0.009	0.864
S4a	0.752	0.203	0.0578	0.020	0.836
S4b	0.755	0.209	0.0576	0.010	0.854
S4c	0.752	0.203	0.0578	0.020	0.836
S5	0.752	0.203	0.0578	0.020	0.836
S6a	0.752	0.203	0.0578	0.020	0.836
S6b	0.752	0.203	0.0578	0.020	0.836

S4a deliberately compares different price bases and is not a preferred means-test specification. S5 found no disagreement between current receipt and preceding-year OPG income among PASS adults. S6a/S6b found no unresolved receipt after PASS; they do not bound the 187 adults whose eligibility remained unresolved. S1c uses applicant age only and retains its stated spouse-age simplification.

Sources: output/phase4/rq1_sensitivity.csv supplies scenario counts, rate estimates and changes; output/phase6/model_sensitivity.csv supplies household counts, fit/reuse status and performance. The S6

household counts come from the latter file because those cells are blank in the former. output/phase6/run_record.json records the S1b/S1d cohort equality. Operational definitions are implemented in analysis/phase4_rq1.py and analysis/phase6_sensitivity.py; their grouped-fold seed is 201415.

Model checks S7–S11

These checks change samples, fitting or evaluation rather than screening rules. S7 refits complete cases; S8, S9 and S11 refit the primary cohort. S10 reweights the primary predictions without refitting. S9 uses person weights as frequency weights when fitting, with unweighted imputation and unweighted held-out evaluation; this is not full survey-design regression. S11 retains five folds while grouping by derived primary sampling unit.

Study group and fitting

ID	Check	Fit	Households	N	Events
S7	Complete cases for schooling and office distance	Refit	811	992	63
S8	Married indicator replaces welfare access	Refit	827	1,013	67
S9	Frequency weights when fitting; unweighted imputation and evaluation	Refit	827	1,013	67
S10	Person-weighted measures of the primary predictions	Reweighted	827	1,013	67
S11	Folds grouped by derived primary sampling unit	Refit	827	1,013	67

Held-out performance (same scenario IDs)

ID	AUC	AP	Brier	CITL	Slope
S7	0.758	0.211	0.0556	−0.029	0.819
S8	0.755	0.207	0.0578	0.014	0.846
S9	0.750	0.193	0.0583	−0.032	0.718
S10	0.763	0.222	0.0641	0.078	0.935
S11	0.758	0.204	0.0575	0.003	0.860

Source: output/phase6/model_sensitivity.csv, rows S7–S11. All refitted checks converged in five of five folds. The S10 performance measures are person-weighted; the other rows use unweighted performance.

Supplementary Table S2. Ridge comparison of settlement coding

Both codings use the same 1,013 adults, 67 events and five primary household-grouped outer folds (seed 201415). The reference is the retained, unpenalised six-slope logistic model. Ridge regression compares the merged six-slope coding with a separate rural-formal contrast requiring seven slopes. The comparison keeps participants and outer folds fixed rather than deleting the sparsely supported category.

Within each outer training set, five inner folds keep households intact and stratify households by whether any member reports non-receipt. Inner seeds equal 201415 plus the outer-fold number. Each inner training fold supplies its own age median, categorical modes and predictor means and population standard deviations. Mode ties select the smallest code. These statistics are applied to that inner validation fold. A zero standard deviation is replaced by one. After tuning, preprocessing is relearned from the complete outer training set before predicting its held-out households. No inner or outer validation observations supply imputation or scaling statistics.

The inverse-penalty grid is C = 0.01, 0.1, 0.3, 1, 3, 10, 30 and 100; smaller C implies stronger shrinkage. The selected value minimises the unweighted mean of the five inner-fold Brier scores. Exact ties select the smallest C. The estimator uses an L2 penalty, an unpenalised intercept, the lbfgs solver and a maximum of 1,000 iterations. Scaling applies to ridge predictors only. The coding decision D-052 stays fixed, so this sensitivity does not remove the outcome information used to combine settlement categories or establish independent validation of that decision.

Coding	Estimator	AUC	AP	Brier	CITL	Slope	Converged outer folds
Merged settlement, six slopes	Ordinary logistic reference	0.752	0.203	0.0578	0.020	0.836	5/5
Merged settlement, six slopes	Ridge, inner-fold tuned	0.756	0.205	0.0576	0.013	0.994	5/5
Separate rural-formal contrast, seven slopes	Ridge, inner-fold tuned	0.750	0.205	0.0577	0.010	0.978	5/5

Outer-fold penalty and convergence record

Coding	Outer fold	Selected C	Converged	Iterations	Inner folds complete
merged six slope	1	0.1	Yes	8	Yes
merged six slope	2	0.1	Yes	8	Yes
merged six slope	3	0.1	Yes	8	Yes
merged six slope	4	0.1	Yes	8	Yes
merged six slope	5	0.1	Yes	8	Yes
separate seven slope	1	0.1	Yes	8	Yes
separate seven slope	2	0.1	Yes	8	Yes
separate seven slope	3	0.1	Yes	8	Yes
separate seven slope	4	0.1	Yes	10	Yes
separate seven slope	5	0.1	Yes	8	Yes

All 400 inner fits and all 10 outer fits converged. The 400 candidate/fold scores and 80 candidate/outer-fold mean scores are retained in inner_fold_scores.csv and inner_grid_scores.csv, respectively. Every outer fold selected C = 0.1.

Sources: output/phase5_ridge_settlement/metrics.csv, fold_penalties.csv, inner_fold_scores.csv, inner_grid_scores.csv and run_record.json; implementation: analysis/phase5_ridge_settlement.py. The earlier seven-slope maximum-

likelihood attempt produced diagnostic AUC 0.739 and calibration slope 0.673, but only four of five folds converged and its unsupported coefficient was approximately −21.88. These failed-attempt summaries are retained in output/phase5_original_7slope/metrics.csv and fold_coefficients.csv; they do not provide a valid superiority ranking against the converged six-slope reference (AUC 0.752; slope 0.836). Small differences between the ridge codings also do not establish superior generalisable performance from merging.

Supplementary Table S3. Twenty repeated household-grouped refits

Twenty consecutive seeds, 201415–201434, repeat the five-fold household-grouped validation. Each repetition relearns imputation and refits the six-slope model inside its training folds. Settlement decision D-052 and the predictor specification remain fixed. Every repetition uses the same 1,013 adults and 67 events. Seed 201415 reproduces the primary performance.

The retained record contains all 20 requested repetitions: 100 fits were attempted, 100 converged and none was reported non-convergent. Each adult has one held-out prediction within each repetition. Repetitions share participants; their spread describes sensitivity to the partition, imputation and refitting under the fixed coding. It is not a 95% confidence interval, adds no independent events and does not repeat the outcome-informed coding decision. Predictions from different repetitions must not be stacked as independent observations.

Panel A. Every repetition

Seed	AUC	AP	Brier	CITL	Slope	Converged folds
201415	0.752	0.203	0.0578	0.020	0.836	5/5
201416	0.733	0.173	0.0588	0.005	0.738	5/5
201417	0.763	0.229	0.0568	0.005	0.904	5/5
201418	0.762	0.234	0.0571	−0.001	0.858	5/5
201419	0.754	0.196	0.0578	−0.014	0.846	5/5
201420	0.749	0.226	0.0574	−0.001	0.838	5/5
201421	0.750	0.199	0.0581	−0.013	0.817	5/5
201422	0.759	0.246	0.0568	0.010	0.893	5/5
201423	0.754	0.214	0.0578	0.015	0.835	5/5
201424	0.746	0.216	0.0582	0.003	0.814	5/5
201425	0.746	0.209	0.0577	0.007	0.805	5/5
201426	0.752	0.199	0.0579	−0.011	0.817	5/5
201427	0.749	0.207	0.0579	−0.002	0.805	5/5
201428	0.757	0.236	0.0569	0.013	0.892	5/5
201429	0.758	0.207	0.0571	0.019	0.887	5/5
201430	0.738	0.177	0.0585	−0.002	0.758	5/5
201431	0.757	0.197	0.0576	0.013	0.843	5/5
201432	0.755	0.220	0.0574	−0.000	0.852	5/5
201433	0.773	0.238	0.0565	−0.004	0.943	5/5
201434	0.743	0.187	0.0584	0.012	0.792	5/5

Panel B. Distribution across 20 repetitions

Measure	Median	Interquartile range	Range
AUC	0.753	0.748–0.757	0.733–0.773
Average precision	0.208	0.199–0.227	0.173–0.246
Brier score	0.0578	0.0571–0.0580	0.0565–0.0588
Calibration-in-the-large	0.004	−0.002–0.012	−0.014–0.020
Calibration slope	0.837	0.812–0.865	0.738–0.943

Panel C. Training-fold coefficient stability

Coefficients are log-odds estimates from the same 100 overlapping training fits. Sign counts and ranges describe stability; they are neither significance tests nor confidence intervals. The manuscript's odds ratios remain the full-cohort model estimates.

Coefficient	Negative fits	Positive fits	Median	Range
Female versus male	100/100	0/100	−0.93	−1.30 to −0.61
Schooling below Grade 7	100/100	0/100	−1.46	−1.97 to −1.14
Age per decade above 60	99/100	1/100	−0.39	−0.81 to 0.01
Urban informal versus urban formal	27/100	73/100	0.16	−0.69 to 0.87
Traditional/rural formal versus urban formal	100/100	0/100	−0.46	−0.99 to −0.02
Office ≥5 km or unavailable	97/100	3/100	−0.41	−0.92 to 0.17

Sources: output/phase5_refit_repeats/repetition_metrics.csv supplies Panel A; summary.csv supplies Panel B; coefficient_summary.csv and fold_coefficients.csv supply Panel C; run_record.json records seeds and fit counters. Implementation: analysis/phase5_refit_repeats.py.

Supplementary Table S4. Descriptive statistics in separate columns

These are the manuscript Table V values before model imputation. N is an observed sample size, not a weighted population count. Age entries are means in completed years. Other unweighted entries are counts and within-receipt-group percentages; weighted entries are person-weighted percentages. Both denominators include missing predictors. Missing/unassigned schooling and missing office-access values are shown explicitly. A recorded office-unavailable category differs from missing office access. Selected categories are displayed; omitted comparison categories remain in their receipt-group denominators. No uncertainty intervals were calculated for these summaries.

Characteristic	Recipients, unweighted (N = 946)	Recipients, weighted	Non-recipients, unweighted (N = 67)	Non-recipients, weighted
Age, mean years	70.4	69.8	67.3	68.4
Female	686 (72.5%)	70.4%	33 (49.3%)	53.4%
Grade 7 or above	290 (30.7%)	33.0%	49 (73.1%)	80.8%
Schooling missing/unassigned	3 (0.3%)	0.4%	0 (0.0%)	0.0%
Urban formal settlement	284 (30.0%)	36.2%	38 (56.7%)	66.2%
Welfare office at least 5 km or unavailable	717 (75.8%)	72.0%	34 (50.7%)	40.3%
Welfare-office access missing	14 (1.5%)	1.8%	4 (6.0%)	7.2%

Source: output/phase4/rq1_characteristics.csv (also retained in output/runs/primary/rq1_characteristics.csv). The source contains three recipients with missing/unassigned schooling, 14 recipients with missing office access and four non-recipients with missing office access. All 67 non-recipients have an assigned schooling category. Percentages describe receipt-group composition, not non-receipt rates, adjusted effects or predictions.

Supplementary Table S5. Recorded literature searches and selection flow

The following details reproduce the author's retained manuscript search record dated 17 August 2026, archived in paper/manuscript.md at Git revision e0b29b9bebb2a573245dc70bf72048da9a7f9b96. They document the reported procedure; no search was rerun to prepare this supplement. Historical search IDs S1–S7, including S1b, identify searches here and are distinct from the sensitivity-scenario IDs in Table S1.

The author record states that Web of Science could not be authenticated through DUT on the search date and Dimensions was used instead. Recorded relevance criteria covered grant access, exclusion, take-up, targeting or classification of household-survey data; English-language peer-reviewed articles, theses and institutional reports; and a preference for South African evidence. The stated empirical-study window was 2015–2026, with foundational theory exempt. Earlier empirical programme studies are also cited in the manuscript, but the retained record does not identify their exception or additional search route. The year window is therefore reported as a recorded search setting, rather than independently verified compliance for every retained or cited source.

Panel A. Sources and counts recorded on 17 August 2026

Historical search ID	Source	Recorded hits	Retained sources
S1	Scopus	43	9
S1b	Scopus, widened	149	10
S2	Scopus, method	100	11
S3	Dimensions	36	5
S4	IEEE Xplore	47	0
S5	SABINET/AJOL	100	9
S6	Google Scholar	35	11
S7	DataFirst/Stats SA	1	0

The eight recorded hit counts sum to 511 and the retained-source counts sum to 55. The author record assigns a repeated retained source to its earliest search. Original exports, record identifiers, a complete 55-source extraction register and item-level screening decisions were not supplied with this record. The allocation of individual sources to searches and the duplicate-removal decisions therefore cannot be independently reproduced from these totals. Recorded hits are historical author-reported counts, not current database result totals.

Panel B. Queries reproduced verbatim from the retained record

Recorded query S1 — Scopus

TITLE-ABS-KEY(("social grant*" OR "social assistance" OR "cash transfer*" OR "social protection") AND (exclusion OR "non-take-up" OR "take-up" OR "coverage gap*" OR "targeting error*" OR "access barrier*") AND ("South Africa*" OR "KwaZulu-Natal"))

Recorded query S1b — Scopus, widened

TITLE-ABS-KEY(("social grant*" OR "social assistance" OR "cash transfer*" OR "social protection" OR "social security" OR SASSA OR "child support grant" OR "social relief of distress" OR SRD) AND (exclusion OR exclud* OR "non-take-up" OR "take-up" OR uptake OR "coverage gap*" OR "targeting error*" OR "access barrier*" OR "non-receipt" OR eligib*) AND ("South Africa*" OR "KwaZulu-Natal"))

Recorded query S2 — Scopus, method

TITLE-ABS-KEY(("machine learning" OR "random forest" OR XGBoost OR "gradient boosting" OR "proxy means test*" OR "predictive model*") AND (poverty OR "social protection" OR "social assistance" OR "cash

transfer*" OR "food insecur*") AND ("household survey" OR "survey data" OR "household data")) AND NOT TITLE-ABS-KEY(cancer OR tumour OR HIV OR malaria OR clinical OR patient*)

Recorded query S3 — Dimensions

("social grant" OR "social assistance" OR "cash transfer" OR "social protection") AND (exclusion OR "take-up" OR "non-receipt" OR "access barriers") AND ("South Africa" OR "KwaZulu-Natal")

Recorded query S4 — IEEE Xplore

("machine learning" OR "classification") AND ("poverty" OR "social welfare" OR "socioeconomic") AND ("survey data")

Recorded query S5 — SABINET/AJOL

social grant AND exclusion AND (KwaZulu-Natal OR "South Africa")

Recorded query S6 — Google Scholar

"social grant" exclusion "South Africa" -site:pinterest.com

Recorded query S7 — DataFirst/Stats SA

Living Conditions Survey 2014/2015 metadata KwaZulu-Natal

Panel C. Reported aggregate selection flow

Stage	Records
Identified by the eight recorded searches	511
Duplicates removed	136
Remaining for title/abstract screening	375
Excluded at title/abstract screening	222
Assessed at full text	153
Excluded at full text	98
Retained in the broader social-grant review	55

The arithmetic is consistent: 511 − 136 = 375; 375 − 222 = 153; and 153 − 98 = 55. These are reported aggregate counts, not an independently rerun selection audit. The 55 sources span several grants and populations; they are not 55 OPG-specific empirical studies. The synthesis compares programme, setting, unit of analysis, outcome definition, findings and limitations. Heterogeneous populations and access stages prevent a pooled OPG non-receipt estimate. The retained query record improves transparency without establishing exhaustive coverage or record-level reproducibility.